

Lecture 1 Preference and Choice

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Individual decision-making under certainty

Objects of inquiry

Our study begins with **individual** decision-making under **certainty**

Items of interest include:

- Feasible set
- Objective function ($\text{Feasible set} \rightarrow \mathbb{R}$)
- Choice correspondence ($\text{Parameters} \Rightarrow \text{Feasible set}$)
- “Maximized” objective function ($\text{Parameters} \rightarrow \mathbb{R}$)

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We start with an even more general problem that only includes

- Feasible set
- Choice correspondence

A fairly innocent assumption will then allow us to treat this model as an optimization problem

Individual decision-making under certainty

Course outline

We will divide decision-making under certainty into three units:

① Producer theory

- Feasible set defined by technology
- Objective function $p \cdot y$ depends on prices

② Abstract choice theory

- Feasible set totally general
- Objective function may not even exist

③ Consumer theory

- Feasible set defined by budget constraint and depends on prices
- Objective function $u(x)$

Outline

- Preferences
 - Preference relations and rationality
 - From preferences to behavior
 - From behavior to preferences: “revealed preference”
- Utility functions
- Properties of preferences
- Behavioral critiques

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Preference relations

Definition (weak preference relation)

$\succsim$ is a binary relation on a set of possible choices X such that
 $x \succsim y$ iff “ x is **at least as good as** y .”

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Definition (strict preference relation)

$\succ$ is a binary relation on X such that $x \succ y$ (“ x is **strictly preferred** to y ”) iff $x \succsim y$ but $y \not\succsim x$.

Definition (indifference)

$\sim$ is a binary relation on X such that $x \sim y$ (“the agent is **indifferent** between x and y ”) iff $x \succsim y$ and $y \succsim x$.

Properties of preference relations

Definition (completeness)

$\succsim$ on X is **complete** iff $\forall x, y \in X$, either $x \succsim y$ or $y \succsim x$.

Completeness implies that $x \succsim x$

Definition (transitivity)

$\succsim$ on X is **transitive** iff whenever $x \succsim y$ and $y \succsim z$, we have $x \succsim z$.

Rules out preference cycles except in the case of indifference

Definition (rationality)

$\succsim$ on X is **rational** iff it is both complete and transitive.

Summary of preference notation

	$y \succsim x$	$y \not\succsim x$
$x \succsim y$	$x \sim y$	$x \succ y$
$x \not\succsim y$	$y \succ x$	Ruled out by completeness

Summary of preference notation

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Can think of (complete) preferences as inducing a function

$$p: X \times X \rightarrow \{\succ, \sim, \prec\}$$

Other properties of rational preference relations

Assume $\succsim$ is rational. Then for all $x, y, z \in X$:

- Weak preference is reflexive: $x \succsim x$
- Indifference is
 - Reflexive: $x \sim x$
 - Transitive: $(x \sim y) \wedge (y \sim z) \implies x \sim z$
 - Symmetric: $x \sim y \iff y \sim x$
- Strict preference is
 - Irreflexive: $x \not\succ x$
 - Transitive: $(x \succ y) \wedge (y \succ z) \implies x \succ z$
- $(x \succ y) \wedge (y \succsim z) \implies x \succ z$, and
 $(x \succsim y) \wedge (y \succ z) \implies x \succ z$

Two strategies for modelling individual decision-making

① Conventional approach

Start from preferences, ask what choices are compatible

② Revealed-preference approach

Start from observed choices, ask what preferences are compatible

- Can we test rational choice theory? How?
- Are choices consistent with maximization of some objective function? Can we recover an objective function?
- How can we use objective function—in particular, do interpersonal comparisons work? If so, how?

Choice rules

Definition (Choice rule)

Given preferences $\succsim$ over X , and choice set $B \subseteq X$, the choice rule is a correspondence giving the set of all “best” elements in B :

$$C(B, \succsim) \equiv \{x \in B : x \succsim y \text{ for all } y \in B\}.$$

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Theorem

Suppose $\succsim$ is complete and transitive and B finite and non-empty. Then $C(B, \succsim) \neq \emptyset$.

Proof of non-emptiness of choice correspondence

Proof.

Proof by mathematical induction on the number of elements in B .

Consider $|B| = 1$ so $B = \{x\}$; by completeness $x \succsim x$, so $x \in C(B, \succsim) \implies C(B, \succsim) \neq \emptyset$.

Suppose that for all $|B| = n \geq 1$, we have $C(B, \succsim) \neq \emptyset$. Consider A such that $|A| = n + 1$; thus $A = B \cup \{x\}$. We can consider some $y \in C(B, \succsim)$ by the inductive hypothesis. By completeness, either

- ① $y \succ x$, in which case $y \in C(A, \succsim)$.
- ② $x \succ y$, in which case $x \in C(A, \succsim)$ by transitivity.

Thus $C(A, \succsim) \neq \emptyset$

The inductive hypothesis holds for all finite n .



Revealed preference

- Before, we used a known preference relation $\succsim$ to generate choice rule $C(\cdot, \succsim)$
- Now we suppose the agent **reveals her preferences** through her choices, which we observe; can we deduce a rational preference relation that could have generated them?

Definition (revealed preference choice rule)

Any $C_R: 2^X \rightarrow 2^X$ (where 2^X means the set of subsets of X) such that for all $A \subseteq X$, we have $C_R(A) \subseteq A$.

If $C_R(\cdot)$ could be generated by a rational preference relation (i.e., there exists some complete, transitive $\succsim$ such that $C_R(A) = C(A, \succsim)$ for all A), we say it is **rationalizable**

Examples of revealed preference choice rules

Suppose we know $C_R(\cdot)$ for

- $A \equiv \{a, b\}$
- $B \equiv \{a, b, c\}$

$C_R(\{a, b\})$	$C_R(\{a, b, c\})$	Possibly rationalizable?
$\{a\}$	$\{c\}$	
$\{a\}$	$\{a\}$	
$\{a, b\}$	$\{c\}$	

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A necessary condition for rationalizability

Suppose that $C_R(\cdot)$ is rationalizable (in particular, it is generated by $\succsim$), and we observe $C_R(A)$ for some $A \subseteq X$ such that

- $a \in C_R(A)$ (a was chosen $\iff a \succsim z$ for all $z \in A$)
- $b \in A$ (b could have been chosen)

We can infer that $a \succ b$

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Now consider some $B \subseteq X$ such that

- $a \in B$
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We can infer that $b \succ a$

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We can infer that $b \succsim a$

Thus $a \sim b$

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We can infer that $b \succsim a$

Thus $a \sim b$, hence $a \in C_R(B)$ and $b \in C_R(A)$ by transitivity

Houthaker's Axiom of Revealed Preferences

A rationalizable choice rule $C_R(\cdot)$ must therefore satisfy “HARP”:

Definition (Houthaker's Axiom of Revealed Preferences)

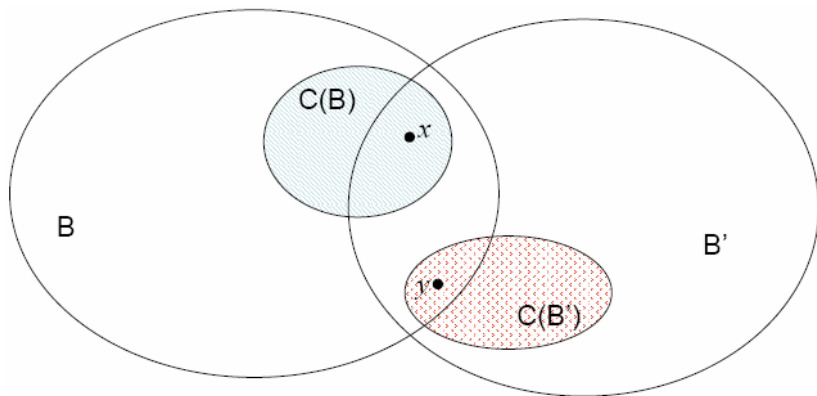
Revealed preferences $C_R: 2^X \rightarrow 2^X$ satisfies HARP iff $\forall a, b \in X$ and $\forall A, B \subseteq X$ such that

- $\{a, b\} \subseteq A$ and $a \in C_R(A)$; and
- $\{a, b\} \subseteq B$ and $b \in C_R(B)$,

we have that $a \in C_R(B)$ (and $b \in C_R(A)$).

Illustrating HARP

A violation of HARP:



Example of HARP

Suppose

- ① Revealed preferences $C_R(\cdot)$ satisfy HARP, and that
 - ② $C_R(\cdot)$ is nonempty-valued (except for $C_R(\emptyset)$)
- If $C_R(\{a, b\}) = \{b\}$, what can we conclude about $C_R(\{a, b, c\})$?
 - If $C_R(\{a, b, c\}) = \{b\}$, what can we conclude about $C_R(\{a, b\})$?

Example of HARP

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$$C_R(\{a, b, c\}) \in \{\{b\}, \{c\}, \{b, c\}\}$$

- If $C_R(\{a, b, c\}) = \{b\}$, what can we conclude about $C_R(\{a, b\})$?

Example of HARP

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- If $C_R(\{a, b, c\}) = \{b\}$, what can we conclude about $C_R(\{a, b\})$?

$$C_R(\{a, b\}) = \{b\}$$

HARP is necessary and sufficient for rationalizability I

Theorem

Suppose revealed preference choice rule $C_R: 2^X \rightarrow 2^X$ is nonempty-valued (except for $C_R(\emptyset)$). Then $C_R(\cdot)$ satisfies HARP iff there exists a rational preference relation $\succsim$ such that $C_R(\cdot) = C(\cdot, \succsim)$.

Proof.

Rationalizability $\Rightarrow$ HARP as argued above.

Rationalizability $\Leftarrow$ HARP: suppose $C_R(\cdot)$ satisfies HARP, we will construct a “revealed preference relation” $\succsim_c$ that generates $C_R(\cdot)$. For any x and y , let $x \succsim_c y$ iff there exists some $A \subseteq X$ such that $y \in A$ and $x \in C_R(A)$.

We must show that $\succsim_c$ is complete, transitive, and generates C (i.e., $C_R(\cdot) = C(\cdot, \succsim_c)$).

HARP is necessary and sufficient for rationalizability II

Proof (continued).

- ① $C_R(\cdot)$ is nonempty-valued, so either $x \in C_R(\{x, y\})$ or $y \in C_R(\{x, y\})$. Thus either $x \succsim_c y$ or $y \succsim_c x$.
- ② Suppose $x \succsim_c y \succsim_c z$ and consider $C_R(\{x, y, z\}) \neq \emptyset$. Thus one (or more) of the following must hold:
 - ① $x \in C_R(\{x, y, z\}) \implies x \succsim_c z$.
 - ② $y \in C_R(\{x, y, z\})$.
But $x \succsim_c y$, so by HARP $x \in C_R(\{x, y, z\}) \implies x \succsim_c z$ by 1.
 - ③ $z \in C_R(\{x, y, z\})$.
But $y \succsim_c z$, so by HARP $y \in C_R(\{x, y, z\}) \implies x \succsim_c z$ by 2.
- ③ We must show that $x \in C_R(B)$ iff $x \succsim_c y$ for all $y \in B$.
 - $x \in C_R(B) \implies x \succsim_c y$ by construction of $\succsim_c$.
 - By nonempty-valuedness, there must be some $y \in C_R(B)$; by HARP, $x \succsim_c y$ implies that $x \in C_R(B)$. □

Revealed preference and limited data

Our discussion relies on **all** preferences being observed

- All elements of $C_R(A)$
- $C_R(A)$ for every $A \subseteq X$

Revealed preference and limited data

Our discussion relies on **all** preferences being observed. . . real data is typically more limited

- All elements of $C_R(A)$. . . we may only see *one* element of A
i.e., $\tilde{C}_R(A) \in C_R(A)$
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i.e., $\hat{C}_R(A): \mathcal{B} \rightarrow 2^X$ for $\mathcal{B} \subset 2^X$ with $\hat{C}_R(A) = C_R(A)$

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Other “axioms of revealed preference” hold in these environments

- Weak Axiom of Revealed Preference (WARP)
- Generalized Axiom of Revealed Preference (GARP)—necessary and sufficient condition for rationalizability

Weak Axiom of Revealed Preferences I

Definition (Weak Axiom of Revealed Preferences)

Revealed preferences $\hat{C}_R: \mathcal{B} \rightarrow 2^X$ defined only for choice sets $\mathcal{B} \subseteq 2^X$ satisfies WARP iff $\forall a, b \in X$ and $\forall A, B \in \mathcal{B}$ such that

- $\{a, b\} \subseteq A$ and $a \in \hat{C}_R(A)$; and
- $\{a, b\} \subseteq B$ and $b \in \hat{C}_R(B)$,

we have that $a \in \hat{C}_R(B)$ (and $b \in \hat{C}_R(A)$).

HARP is WARP with all possible choice sets (i.e., $\mathcal{B} = 2^X$)

WARP is **necessary** but **not sufficient** for rationalizability

Weak Axiom of Revealed Preferences II

WARP is **not sufficient** for rationalizability

Example

Consider $\hat{C}_R: \mathcal{B} \rightarrow 2^{\{a,b,c\}}$ defined for choice sets $\mathcal{B} \equiv \{\{a, b\}, \{b, c\}, \{c, a\}\} \subseteq 2^{\{a,b,c\}}$ with:

- $\hat{C}_R(\{a, b\}) = \{a\},$
- $\hat{C}_R(\{b, c\}) = \{b\},$ and
- $\hat{C}_R(\{c, a\}) = \{c\}.$

$\hat{C}_R(\cdot)$ **satisfies WARP**, but is **not rationalizable**.

Think of $\hat{C}_R(\cdot)$ as a restriction of some $C_R: 2^{\{a,b,c\}} \rightarrow 2^{\{a,b,c\}};$ there is no $C_R(\{a, b, c\})$ consistent with HARP

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 - From preferences to behavior
 - From behavior to preferences: “revealed preference”
- Utility functions
- Properties of preferences
- Behavioral critiques

From abstract preferences to maximization

- Our model of choice so far is entirely abstract
- Utility assigns a numerical ranking to each possible choice
- By assigning a utility to each element of X , we turn the choice problem into an optimization problem

Definition (utility function)

Utility function $u: X \rightarrow \mathbb{R}$ represents $\succsim$ on X iff for all $x, y \in X$,

$$x \succsim y \iff u(x) \geq u(y).$$

Then the choice rule is

$$C(B, \succsim) \equiv \{x \in B: x \succsim y \text{ for all } y \in B\} = \operatorname{argmax}_{x \in B} u(x)$$

Utility representation implies rationality

Theorem

If utility function $u: X \rightarrow \mathbb{R}$ represents $\succsim$ on X , then $\succsim$ is rational.

Proof.

For any $x, y \in X$, we have $u(x), u(y) \in \mathbb{R}$, so either $u(x) \geq u(y)$ or $u(y) \geq u(x)$. Since $u(\cdot)$ represents $\succsim$, either $x \succsim y$ or $y \succsim x$; i.e., $\succsim$ is complete.

Suppose $x \succsim y \succsim z$. Since $u(\cdot)$ represents $\succsim$, we know $u(x) \geq u(y) \geq u(z)$. Thus $u(x) \geq u(z) \implies x \succsim z$. $\succsim$ is transitive. □

Ordinality of utility and interpersonal comparisons

Note that $\succsim$ is represented by **any** function satisfying

$$x \succsim y \iff u(x) \geq u(y)$$

for all $x, y \in X$

Thus **any increasing monotone transformation of $u(\cdot)$ also represents $\succsim$**

- The property of representing $\succsim$ is **ordinal**
- There is **no such thing as a “util”**

Failure of interpersonal comparisons

Interpersonal comparisons are impossible using this theory

- ❶ Disappointing to original utilitarian agenda
- ❷ Rawls (following Kant, following. . .) attempts to solve this by asking us to consider only a single chooser
- ❸ “Just noticeable difference” suggests defining one util as the smallest difference an individual can notice
 - $x \succsim y$ iff $u(x) \geq u(y) - 1$
 - Note $\succ$ is transitive, but $\succsim$ is not

Can we find a utility function representing $\succsim$? I

Theorem

Any *complete* and *transitive* preference relation $\succsim$ on a *finite set* X can be represented by some utility function $u: X \rightarrow \{1, \dots, n\}$ where $n \equiv |X|$.

Intuitive argument:

- ① Assign the “top” elements of X utility $n = |X|$
- ② Discard them; we are left with a set X'
- ③ If $X' = \emptyset$, we are done; otherwise return to step 1 with the set X'

Can we find a utility function representing $\succsim$? II

Proof.

Proof by mathematical induction on $n \equiv |X|$. The theorem holds trivially for $n = 0$, since $X = \emptyset$.

Suppose the theorem holds for sets with at most n elements.

Consider a set X with $n + 1$ elements. $C(X, \succsim) \neq \emptyset$, so

$Y \equiv X \setminus C(X, \succsim)$ has at most n elements. By inductive hypothesis, preferences on Y are represented by some $u: Y \rightarrow \{1, \dots, n\}$. We extend u to X by setting $u(x) = n + 1$ for all $x \in C(X, \succsim)$.

We must show that this extended u represents $\succsim$ on X .

Can we find a utility function representing $\succsim$? III

Proof (continued).

We must show that this extended u represents $\succsim$ on X ; i.e., that for all x and $y \in X$, we have $x \succsim y$ iff $u(x) \geq u(y)$.

- $x \succsim y \implies u(x) \geq u(y)$. If $x \in C(X, \succsim)$, then $u(x) = n + 1 \geq u(y)$. If $x \notin C(X, \succsim)$, then by transitivity $y \notin C(X, \succsim)$, so x and $y \in Y$. Since u represents $\succsim$ on Y , we must have $u(x) \geq u(y)$.
- $x \succsim y \iff u(x) \geq u(y)$. If $u(x) = n + 1$, then $x \in C(X, \succsim)$, hence $x \succsim y$. If $u(x) \leq n$, then x and $y \in Y$. Since u represents $\succsim$ on Y , we must have $x \succsim y$.

Thus the inductive hypothesis holds for all finite n . □

What if $|X| = \infty$? I

If X is infinite, our proof doesn't go through, but we still *may* be able to represent $\succsim$ by a utility function

Example

Preferences over $\mathbb{R}_+$ with $x_1 \succsim x_2$ iff $x_1 \geq x_2$.

$\succsim$ can be represented by $u(x) = x$. (It can also be represented by other utility functions.)

What if $|X| = \infty$? II

However, if X is infinite we can't necessarily represent $\succsim$ by a utility function

Example (lexicographic preferences)

Preferences over $[0, 1]^2 \subseteq \mathbb{R}^2$ with $(x_1, y_1) \succsim (x_2, y_2)$ iff

- $x_1 > x_2$, or
- $x_1 = x_2$ and $y_1 \geq y_2$.

Lexicographic preferences can't be represented by a utility function

- There are no indifference curves
- A utility function would have to be an order-preserving one-to-one mapping from the unit square to the real line (both are infinite, but they are “different infinities”)

Continuous preferences I

Definition (continuous preference relation)

A preference relation $\succsim$ on X is **continuous** iff for any sequence $\{(x_n, y_n)\}_{n=1}^{\infty}$ with $x_n \succsim y_n$ for all n ,

$$\lim_{n \rightarrow \infty} x_n \succsim \lim_{n \rightarrow \infty} y_n.$$

Equivalently, $\succsim$ is continuous iff for all $x \in X$, the upper and lower contour sets of x

$$\text{UCS}(x) \equiv \{\xi \in X : \xi \succsim x\}$$

$$\text{LCS}(x) \equiv \{\xi \in X : x \succsim \xi\}$$

are both closed sets.

Continuous preferences II

Theorem

A *continuous, rational* preference relation $\succsim$ on $X \subseteq \mathbb{R}^n$ can be represented by a continuous utility function $u: X \rightarrow \mathbb{R}$.

(Note it may also be represented by noncontinuous utility functions)

Full proof in Debron and MWG; abbreviated proof in notes

Outline

- Preferences
 - Preference relations and rationality
 - From preferences to behavior
 - From behavior to preferences: “revealed preference”
- Utility functions
- Properties of preferences
- Behavioral critiques

Reasons for restricting preferences

Analytical tractability often demands restricting “allowable” preferences

- Some restrictions are mathematical conveniences and cannot be empirically falsified (e.g., continuity)
- Some hold broadly (e.g., monotonicity)
- Some require situational justification

Restrictions on preferences imply restrictions on utility functions

Reasons for restricting preferences

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Restrictions on preferences imply restrictions on utility functions

Assumptions for the rest of this section

- 1 $\succsim$ is **rational** (i.e., complete and transitive).
- 2 For simplicity, we assume preferences over $X \subseteq \mathbb{R}^n$.

Properties of rational $\succsim$: Nonsatiation

Definition (monotonicity)

$\succsim$ is **monotone** iff $x \succ y \implies x \succsim y$. (N.B. MWG differs)

$\succsim$ is **strictly monotone** iff $x \succ y \implies x \succ y$.

i.e., more of something is (strictly) better

Definition (local non-satiation)

$\succsim$ is **locally non-satiated** iff for any y and $\varepsilon > 0$, there exists x such that $\|x - y\| \leq \varepsilon$ and $x \succ y$.

implies there are no “thick” indifference curves

$\succsim$ is locally non-satiated iff $u(\cdot)$ has no local maxima in X

Properties of rational $\succsim$: Convexity I

Convex preferences capture the idea that **agents like diversity**

- ① Satisfying in some ways: rather alternate between juice and soda than have either one every day
- ② Unsatisfying in others: rather have a glass of either one than a mixture
- ③ Key question is granularity of goods aggregation
 - Over time? What period?
 - Over what “bite size”?

Properties of rational $\succsim$: Convexity II

Definition (convexity)

$\succsim$ is **convex** iff $x \succsim y$ and $x' \succsim y$ together imply that

$$tx + (1 - t)x' \succsim y \text{ for all } t \in (0, 1).$$

Equivalently, $\succsim$ is convex iff the upper contour set of any y (i.e., $\{x \in X : x \succsim y\}$) is a convex set.

$\succsim$ is **strictly convex** iff $x \succsim y$ and $x' \succsim y$ (with $x \neq x'$) together imply that

$$tx + (1 - t)x' \succ y \text{ for all } t \in (0, 1).$$

i.e., one never gets worse off by mixing goods

$\succsim$ is (strictly) convex iff $u(\cdot)$ is (strictly) quasiconcave

► Definition

Implications for utility representation

Property of $\succsim$	Property of $u(\cdot)$
Monotone	Nondecreasing
Strictly monotone	Increasing
Locally non-satiated	Has no local maxima in X
Convex	Quasiconcave
Strictly convex	Strictly quasiconcave

Properties of rational $\succsim$: Homotheticity

Definition (homotheticity)

$\succsim$ is **homothetic** iff for all x, y , and all $\lambda > 0$,

$$x \succsim y \iff \lambda x \succsim \lambda y.$$

Continuous, strictly monotone $\succsim$ is homothetic iff it can be represented by a utility function that is

Properties of rational $\succsim$: Homotheticity

Definition (homotheticity)

$\succsim$ is **homothetic** iff for all x, y , and all $\lambda > 0$,

$$x \succsim y \iff \lambda x \succsim \lambda y.$$

Continuous, strictly monotone $\succsim$ is homothetic iff it can be represented by a utility function that is homogeneous of degree one (note it can also be represented by utility functions that aren't)

Properties of rational $\succsim$: Quasi-linearity I

Suppose rational $\succsim$ over $X \equiv \mathbb{R} \times Y$

- First good is the numeraire (a.k.a. “good zero” or “good one,” confusingly): think money
- Other goods general; need not be in $\mathbb{R}^n$

Theorem

Suppose rational $\succsim$ on $X \equiv \mathbb{R} \times Y$ satisfies the “numeraire properties”:

- 1 **Good 1 is valuable:** $(t, y) \succsim (t', y) \iff t \geq t'$ for all y ;
- 2 **Compensation is possible:** For every $y, y' \in Y$, there exists some $t \in \mathbb{R}$ such that $(0, y) \sim (t, y')$;
- 3 **No wealth effects:** If $(t, y) \succsim (t', y')$, then for all $d \in \mathbb{R}$, $(t + d, y) \succsim (t' + d, y')$.

...

Properties of rational $\succsim$: Quasi-linearity II

Theorem (continued.)

Then there exists a utility function representing $\succsim$ of the form $u(t, y) = t + v(y)$ for some $v: Y \rightarrow \mathbb{R}$. (Note it can also be represented by utility functions that aren't of this form.)

Conversely, any $\succsim$ on $X = \mathbb{R} \times Y$ represented by a utility function of the form $u(t, y) = t + v(y)$ satisfies the above properties.

Properties of rational $\succsim$: Quasi-linearity III

Proof.

Suppose the numeraire properties hold. Fix some $\bar{y} \in Y$. Define a function $v: Y \rightarrow \mathbb{R}$ such that $(0, y) \sim (v(y), \bar{y})$; this is possible by condition 2.

By condition 3, for any (t, y) and (t', y') , we have $(t, y) \sim (t + v(y), \bar{y})$ and $(t', y') \sim (t' + v(y'), \bar{y})$. Thus $(t, y) \succsim (t', y')$ iff $(t + v(y), \bar{y}) \succsim (t' + v(y'), \bar{y})$ (by transitivity), which holds by condition 1 iff $t + v(y) \geq t' + v(y')$.

The converse is trivial.



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Problems with rational choice

Rational choice theory plays a central role in most tools of economic analysis

But... significant research calls into question underlying assumptions, identifying and explaining deviations using

- Psychology
- Sociology
- Cognitive neuroscience (“neuroeconomics”)

Context-dependent choice

Choices appear to be highly situational, depending on

- 1 Other available options
- 2 Way that options are “framed”
- 3 Social context/emotional state

Numerous research projects consider these effects in real-world and laboratory settings

Non-considered choice

Rational choice theory depends on a *considered* comparison of options

- Pairwise comparison
- Utility maximization

Many actual choices appear to be made using

- ① Intuitive reasoning
- ② Heuristics
- ③ Instinctive desire